ME5160 Advanced Dynamics

Final Project

DDP/iLQR, MPC, Reinforcement Learning, and Generative Trajectory Optimization

Dr. Pilwon Hur
Mechanical and Robotics Engineering, GIST

Assigned: June 14, 2026 | Report & Presentation Due: June 30, 2026

Project Overview and Grading

This **Final Project** replaces a separate Homework 04: its content is folded into the project. It is **worth 150 points** and is intentionally more substantial than HW01–HW03.

- **Project Report (120 pts):** four analytical/computational problems (**Part A**, $4 \times 20 = 80$ pts), simulation *with animation* on two self-selected problems (**Part B**, 30 pts), and report quality & cross-topic synthesis (**Part C**, 10 pts).
- **Project Presentation (30 pts):** a **pre-recorded video presentation (under 10 minutes)** with slides and an animation/demo (see the final section).
- Each of the four problems maps to one lecture: **P1** $\rightarrow$ DDP/iLQR (Lec. 11), **P2** $\rightarrow$ MPC (Lec. 12), **P3** $\rightarrow$ RL (Lec. 13), **P4** $\rightarrow$ Generative TO (Lec. 14).
- Each problem extends a canonical benchmark with one new ingredient: *actuator dynamics, state/input constraints, a non-holonomic constraint,* or *multimodality.*
- **All four problems must be answered and explained** (Part A). Show all work; explain your reasoning, assumptions, and notation clearly. Where equations of motion are needed they are a **prerequisite step and are not separately graded**. Only the *simulation* component (Part B) is self-selected (two of the four).
- Any programming language is allowed for simulation. Provide code and instructions to reproduce all figures and animations.
- References: Tedrake, *Underactuated Robotics*; Rawlings, Mayne & Diehl, *Model Predictive Control*; Sutton & Barto, *Reinforcement Learning*; Chi et al., *Diffusion Policy*; Lipman et al., *Flow Matching*.
- **Submission:** upload a single `.zip` archive containing (i) the report as **PDF**, (ii) all **script/code files**, (iii) the presentation file (**`.pptx` or `.pdf`**), and (iv) the **recorded presentation video (`.mp4`)**.

Parameters: what is given vs. what you choose

Use the **standard physical parameters** below so derivations and simulations are well defined and comparable. You must **choose and justify the design/tuning parameters** yourself: cost weights (e.g. $\mathbf{Q}, \mathbf{R}, q_\theta, q_\omega, r$), horizon N, time step h or T_s, terminal penalty, input/state limits ($u_{\max}, x_{\max}, \tau_{\max}$), discount γ, and learning hyperparameters. State every value you use in your report.

System	Physical parameter	Suggested value
Cart-pole (P1, P2)	cart mass m_c	1.0 kg
	pole mass m_p	0.2 kg
	pole length ℓ	0.5 m
	gravity g	9.81 m/s^2
	actuator time constant τ_a (P1)	0.05 s
Mobile inverted pendulum (P3)	body mass m_b	0.5 kg
	body CoM height ℓ	0.3 m
	wheel radius R	0.05 m
	wheel mass m_w	0.05 kg
Planar navigation (P4)	workspace	$[0, 10] \times [0, 10]$
	obstacle centers p_o	$(4, 6),\ (7, 3)$
	obstacle radii (inflated) r	$1.2,\ 1.0$

These are reasonable defaults, not the only valid choices: you may use a different (justified) parameter set, and the planar scene in P4 may be redesigned to expose multiple homotopy classes.

1 Cart-Pole with Actuator Dynamics: DDP and iLQR (20 points)

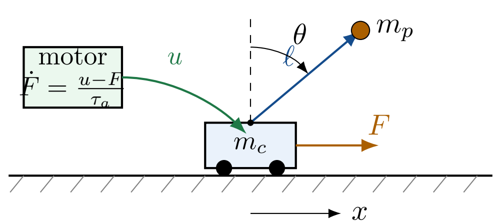

Figure 1: Cart-pole with first-order actuator dynamics. The commanded input u drives an actual force F on the cart through a motor lag $\dot{F} = (u - F)/\tau_a$. State: $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}, F]^\top$. Pole angle θ measured from the downward vertical; upright is $\theta = \pi$.

Consider the cart-pole of Figure 1: a cart of mass m_c on a horizontal track carrying a pole (point mass m_p, length ℓ). Unlike HW01–HW03, the horizontal force is produced by an actuator with first-order lag, so the **commanded** input u and the **actual** force F differ: $\dot{F} = (u - F)/\tau_a$.

Prerequisite (not graded). Derive the cart-pole equations of motion in manipulator form $\mathbf{M}(\theta)\ddot{\mathbf{q}} + \mathbf{C}(\theta, \dot{\theta})\dot{\mathbf{q}} + \mathbf{g}(\theta) = \mathbf{B}_q F$ with $\mathbf{q} = [x, \theta]^\top$, convert to first-order form, and append the actuator state F.

(a) **(2 pts)** Write the augmented continuous-time dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, u)$ with $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}, F]^\top$. Discretize with forward Euler of step h (and note RK4 as an alternative) to obtain the discrete map $\mathbf{x}_{k+1} = F_d(\mathbf{x}_k, u_k)$. Define a nominal trajectory $\{\bar{\mathbf{x}}_k, \bar{u}_k\}$ and the deviations $\delta\mathbf{x}_k = \mathbf{x}_k - \bar{\mathbf{x}}_k$, $\delta u_k = u_k - \bar{u}_k$.

(b) **(2 pts)** For a finite horizon N write the swing-up cost $J = \ell_N(\mathbf{x}_N) + \sum_{k=0}^{N-1} \ell_k(\mathbf{x}_k, u_k)$ and the Bellman recursion $V_k(\mathbf{x}) = \min_u \left[\ell_k(\mathbf{x}, u) + V_{k+1}(F_d(\mathbf{x}, u))\right]$. Define the local action-value function $Q_k(\delta\mathbf{x}, \delta u)$ as the change in $\ell_k + V_{k+1}$ about $(\bar{\mathbf{x}}_k, \bar{u}_k)$.

(c) **(3 pts)** Derive the quadratic-expansion blocks

$$Q_x = \ell_x + \mathbf{A}_k^\top V_x', \qquad Q_u = \ell_u + \mathbf{B}_k^\top V_x',$$
$$Q_{xx} = \ell_{xx} + \mathbf{A}_k^\top V_{xx}' \mathbf{A}_k, \qquad Q_{uu} = \ell_{uu} + \mathbf{B}_k^\top V_{xx}' \mathbf{B}_k, \qquad Q_{ux} = \ell_{ux} + \mathbf{B}_k^\top V_{xx}' \mathbf{A}_k,$$

where $\mathbf{A}_k = \partial F_d / \partial \mathbf{x}$, $\mathbf{B}_k = \partial F_d / \partial u$, and V_x', V_{xx}' are evaluated at step $k+1$. Identify precisely which additional terms appear in Q_{xx}, Q_{uu}, Q_{ux} for **full DDP** (the dynamics second-derivative tensors contracted with V_x').

(d) **(3 pts)** Minimizing the quadratic Q_k over δu, derive the feedforward $\mathbf{k} = -Q_{uu}^{-1} Q_u$ and feedback gain $\mathbf{K} = -Q_{uu}^{-1} Q_{ux}$, and the resulting value-function recursion

$$V_x = Q_x - \mathbf{K}^\top Q_{uu} \mathbf{k}, \qquad V_{xx} = Q_{xx} - \mathbf{K}^\top Q_{uu} \mathbf{K}.$$

State the terminal condition (V_x, V_{xx}) at $k = N$ and write the backward pass as a step-by-step algorithm.

(e) **(3 pts) iLQR vs. DDP.** State exactly which terms iLQR omits. For this system the only nonzero dynamics second derivative is the pole-angle curvature $\partial^2 f^{(\dot{\theta})} / \partial \theta^2 \propto \sin\theta$. Compute the resulting DDP-only contribution $\sum_i V_x'^{(i)} f_{xx}^{(i)}$ to Q_{xx} and discuss when retaining it is worthwhile.

(f) **(3 pts) Forward pass and robustness.** Write the forward rollout with line-search step $\alpha \in (0, 1]$, $\delta u_k = \alpha\, \mathbf{k}_k + \mathbf{K}_k(\mathbf{x}_k - \bar{\mathbf{x}}_k)$, and the predicted cost decrease $\Delta J(\alpha) = \alpha \sum_k Q_u^\top \mathbf{k} + \frac{\alpha^2}{2} \sum_k \mathbf{k}^\top Q_{uu} \mathbf{k}$ used to accept/reject the step. Explain Levenberg–Marquardt regularization $Q_{uu} \leftarrow Q_{uu} + \mu \mathbf{I}$ and why it is needed when $Q_{uu} \not\succ 0$.

(g) **(2 pts) LQR special case.** Show that for linear dynamics and quadratic cost the backward pass reduces *exactly* to the finite-horizon LQR Riccati recursion. Carry out **one scalar backward step** for a scalar linear-quadratic example to demonstrate this.

(h) **(2 pts) Effect of actuator dynamics.** Explain how τ_a affects the conditioning of Q_{uu}, the achievable control bandwidth, and the largest stable step h. Contrast the augmented model with the actuator-free model ($\tau_a \to 0$).

Simulation & Animation option (Tier 1, up to 13 pts — see final section)

Implement the iLQR/DDP backward and forward passes with line search and regularization for swing-up of the cart-pole with actuator dynamics. Plot cost-vs-iteration convergence and the optimized state/control. **Animate** the swing-up. Verify by forward-simulating the optimized controls with a high-accuracy integrator and comparing to the iLQR rollout.

2 Constrained Cart-Pole: Model Predictive Control (20 points)

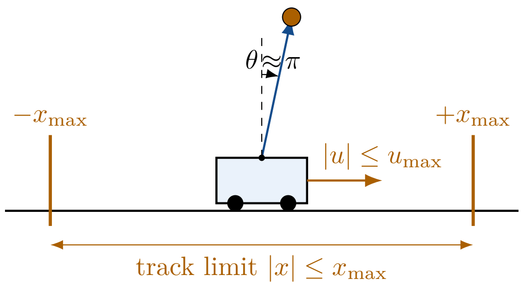

Figure 2: Constrained cart-pole. The cart force saturates, $|u| \le u_{\max}$, and the cart position is bounded by the track, $|x| \le x_{\max}$. The upright equilibrium is $\theta = \pi$, $\dot\theta = 0$.

Consider the cart-pole of Figure 2 with input saturation $|u| \le u_{\max}$ and track limit $|x| \le x_{\max}$. (You may optionally retain the actuator state of Problem 1.)

Prerequisite (not graded). Linearize the dynamics about the upright equilibrium and discretize with step T_s to obtain $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}u_k$.

(a) **(2 pts)** Write the nominal finite-horizon **regulation MPC** problem

$$\min_{\mathbf{x}_i,u_i} \quad \mathbf{x}_N^\top \mathbf{P}\mathbf{x}_N + \sum_{i=0}^{N-1} \left(\mathbf{x}_i^\top \mathbf{Q}\mathbf{x}_i + r u_i^2\right)$$

$$\text{subject to} \quad \mathbf{x}_0 = \hat{\mathbf{x}}_k, \quad \mathbf{x}_{i+1} = \mathbf{A}\mathbf{x}_i + \mathbf{B}u_i, \quad |u_i| \le u_{\max}, \quad |[1\ 0\ 0\ 0]\mathbf{x}_i| \le x_{\max}, \quad \mathbf{x}_N \in \mathcal{X}_f,$$

and state the receding-horizon algorithm (solve, apply $u_0^\star$, shift, re-solve).

(b) **(3 pts) Linear MPC as a QP.** Write the problem as a quadratic program in both the **sparse** form (decision vector stacks all $\mathbf{x}_i$ and u_i, dynamics as equality constraints) and the **condensed** form (eliminate the states so the decision vector is $\mathbf{U} = [u_0, \dots, u_{N-1}]^\top$). Give the condensed Hessian $\mathbf{H}$ and gradient $\mathbf{f}$ explicitly in terms of the prediction matrices.

(c) **(2 pts)** Count the scalar decision variables and the equality/inequality constraints in each form for horizon N, n states, $m = 1$ input. Comment on the sparsity pattern and which form scales better as N grows.

(d) **(2 pts) Feedback by re-solving.** Explain why applying only $u_0^\star$ and re-solving at the next measured state produces an *implicit* feedback law even though no closed-form controller is written. State the relationship to LQR as $N \to \infty$ with terminal cost $\mathbf{P}$ equal to the discrete-algebraic-Riccati solution.

(e) **(3 pts) Recursive feasibility & stability.** State the terminal-cost and terminal-set conditions, give the "shift the previous optimal sequence" argument for recursive feasibility, and sketch the Lyapunov-decrease proof $V^\star(\mathbf{x}_{k+1}) - V^\star(\mathbf{x}_k) \le -\ell(\mathbf{x}_k, u_0^\star)$. Explain the terminal-equality constraint $\mathbf{x}_N = \mathbf{0}$ as the simplest special case.

(f) **(3 pts) Nonlinear MPC swing-up.** Formulate NMPC for swing-up with a wrap-safe stage cost $q_\theta(1 + \cos\theta_i) + q_\omega\omega_i^2 + r u_i^2$. Compare direct multiple shooting vs. direct collocation as the transcription (decision variables, how dynamics are enforced, one pro/con each). Describe warm-starting by trajectory shifting and the **real-time iteration** scheme (one solver step per sample). Connect to using iLQR (Problem 1) as the inner solver.

(g) **(3 pts) Soft constraints.** Replace the hard track limit with slack variables $\sigma_i \geq 0$ and an exact-penalty term $\rho \sum_i \sigma_i$ (ρ large) so the problem stays feasible if the limit is momentarily violated. Write the modified QP. Summarize the **tube-MPC** idea for bounded disturbances in one short paragraph.

(h) **(2 pts) Computational budget.** For sample time T_s and horizon N, estimate the QP size (variables/constraints) and discuss the solve-time-vs-T_s requirement for real-time use and how warm starting helps meet it.

Simulation & Animation option (Tier 1, up to 13 pts)

Implement a receding-horizon MPC loop calling a QP/NLP solver (e.g. OSQP, `quadprog`, CasADi, `scipy.optimize`, `fmincon`). Show constraint satisfaction (track limit and saturation), plot the closed-loop trajectory and the first-input feedback, and **animate**. Report the per-step solve time and demonstrate recursive feasibility empirically.

3 Non-Holonomic Mobile Inverted Pendulum: Reinforcement Learning (20 points)

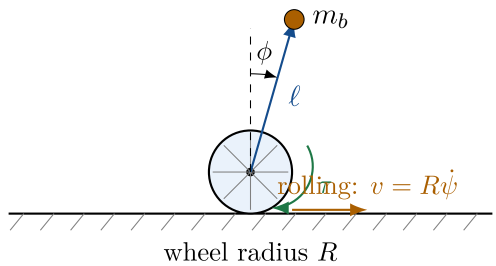

Figure 3: Two-wheeled mobile inverted pendulum (self-balancing robot). The body of mass m_b balances above a driven wheel of radius R; ϕ is the body tilt and τ the wheel torque. The wheels roll without slipping, giving a *non-holonomic* velocity constraint. The system is underactuated and contact-rich.

Consider the mobile inverted pendulum of Figure 3: a body that must balance above a driven wheel while the wheel rolls without slipping (a non-holonomic constraint). The system is underactuated, contact-rich, and parameter-uncertain, which motivates a learned feedback policy rather than a single open-loop plan. (The **acrobot** is an acceptable substitute; if used, replace the non-holonomic constraint with actuator saturation and partial observability.)

Prerequisite (not graded). State the constrained Lagrangian dynamics with the rolling constraint and explain why classical trajectory optimization / MPC is difficult here.

(a) **(3 pts) MDP formulation.** Define the state, the observation (note any partial observability), the action parameterization (wheel torque), the reward (balance + velocity tracking + control effort + smoothness), and the discount factor γ. Justify why the reward is the negative of a running cost.

(b) **(3 pts) Value functions and Bellman.** Define the return $G_k = \sum_{j\geq 0} \gamma^j r_{k+j}$, the value functions $V^\pi(s)$ and $Q^\pi(s, a)$, and write the Bellman expectation and optimality equations. Define the temporal-difference error $\delta_k = r_k + \gamma V(s_{k+1}) - V(s_k)$ and the generalized advantage estimate $\hat{A}_k = \sum_{l\geq 0} (\gamma\lambda)^l \delta_{k+l}$.

(c) **(3 pts) Policy gradient.** Derive the policy-gradient identity

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}\Big[\textstyle\sum_k \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_k \mid s_k)\, \hat{A}_k\Big],$$

interpret it as *stochastic trajectory optimization*, give the causality/advantage form, and write the log-density of a Gaussian policy $\pi_{\boldsymbol{\theta}}(a \mid s) = \mathcal{N}(\mu_{\boldsymbol{\theta}}(s), \Sigma_{\boldsymbol{\theta}}(s))$ for continuous control.

(d) **(3 pts) Algorithm selection.** Contrast on-policy vs. off-policy and value-based vs. policy-based methods. Compare PPO, DDPG/TD3, and SAC, and recommend one for this system, justifying your choice by sample efficiency, exploration, action bounds, and training stability.

(e) **(2 pts) LQR as RL.** Linearize about the upright equilibrium; show the optimal value function is quadratic, $V(\mathbf{x}) = -\mathbf{x}^\top \mathbf{P}\mathbf{x} + c$, and the optimal policy is linear, $\mathbf{u}^\star = -\mathbf{K}\mathbf{x}$ (Riccati). Explain what a learned nonlinear policy adds beyond this local solution.

(f) **(2 pts) Model-based RL and the MPC link.** Describe using a learned dynamics model inside a sampling-based MPC (e.g. MPPI/CEM) and the idea of guided policy search. Discuss model bias and why short-horizon rollouts are used.

(g) **(2 pts) Safety and constraints.** Describe a constrained-MDP / safety-filter / control-barrier-function approach that keeps the body within a safe tilt envelope, and the residual-RL idea of learning a correction on top of a baseline balancing controller.

(h) **(2 pts) Sim-to-real and sample complexity.** Explain domain randomization, curriculum learning, and massively parallel simulation. Estimate the number of environment samples for a PPO run with E parallel environments, T steps per rollout, and I iterations, and discuss the sample-efficiency obstacle.

Simulation & Animation option (Tier 2, up to 15 pts)

Train a policy (PPO/SAC or a clearly documented implementation) for the non-holonomic mobile inverted pendulum (or acrobot). Plot the learning curve (return vs. samples), evaluate from multiple initial conditions, and **animate** a rollout of the learned policy. Verify against a baseline (e.g. LQR near upright) and/or a domain-randomization sweep.

4 Generative Trajectory Optimization: Diffusion and Flow Matching (20 points)

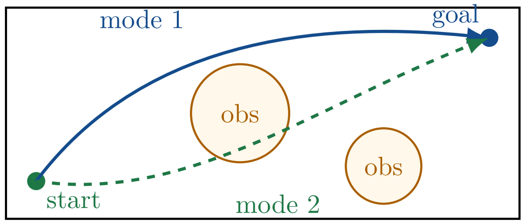

Figure 4: Multimodal planar navigation. Two obstacles create distinct homotopy classes of feasible paths; a generative model represents the *distribution* over trajectories rather than a single local optimum. The cart-pole swing-up (swing-left vs. swing-right) is an analogous low-dimensional multimodal example.

Consider trajectory generation for the planar navigation task of Figure 4, where obstacles create several feasible homotopy classes, and the cart-pole swing-up as a low-dimensional example with two qualitatively different (left/right) local optima.

(a) **(2 pts) Why generative?** Explain how non-convex trajectory optimization produces *multimodal* solution distributions and how local solvers depend on basins of attraction. Give a concrete example of two distinct cart-pole swing-up local optima.

(b) **(3 pts) Diffusion foundations.** For a trajectory/action chunk $A^\star$, write the forward noising $A_t = \sqrt{\bar{\alpha}_t}A^\star + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, the reverse denoiser, and the score interpretation. **Derive the simplified denoising objective $\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(A_t, t, \mathbf{o})\|^2]$** and describe ancestral sampling from noise to a trajectory.

(c) **(3 pts) Diffusion Policy.** Describe the policy as a conditional generative model over an H-step action chunk, receding-horizon execution (sample, execute the first few actions, observe, repeat), why action chunking helps, and the relation to behavior cloning. Explain why executing a full open-loop chunk is risky.

(d) **(3 pts) Cost / constraint guidance.** For a generated planar path $P = [p_0^\top, \ldots, p_N^\top]^\top$ and a circular obstacle of center p_o, inflated radius r, write the penalty $\Phi_{\text{obs}}(P) = \sum_k [\max(0, r - \|p_k - p_o\|)]^2$ and derive its gradient $\nabla_{p_k}\Phi_{\text{obs}} = -2(r - d_k)\frac{p_k - p_o}{d_k}$ for $d_k = \|p_k - p_o\| < r$. Handle the singularity at $p_k = p_o$ with a smoothed norm. Describe guided denoising, a dynamics-residual penalty, and projection-after-denoising.

(e) **(3 pts) Diffusion as a warm-start generator (DiffuSolve).** Explain how diffusion samples seed a non-convex solver (direct collocation or iLQR from Problems 1–2), the basin-coverage interpretation, and why this improves convergence on multimodal problems.

(f) **(3 pts) Flow matching.** State the continuity equation and the conditional flow-matching objective. Using the linear conditional path $z_t = (1-t)z_0 + tz_1$ with target velocity $u_t = z_1 - z_0$, **derive the two-mode 1-D example** (data modes near $z_1 = \pm 1$) and explain how different source samples flow to different modes. Describe sampling by ODE integration.

(g) **(2 pts) Comparison.** Build a table comparing diffusion, flow matching, direct collocation, DDP/iLQR, MPC, and RL on: sampling/compute cost, ability to represent multimodality, constraint handling, feedback vs. open-loop, and data requirements.

(h) **(1 pt) Safety limits.** Discuss the feasibility/safety limitations of learned generative models inside a physical control loop: where learning helps and where physics (dynamics, constraints, feedback) must still dominate.

Simulation & Animation option (Tier 2, up to 15 pts)

Implement diffusion *or* flow-matching trajectory generation for the planar/obstacle task (or cart-pole swing-up). Show multiple generated samples covering distinct modes/homotopy classes, demonstrate cost/obstacle guidance, and **animate** at least one generated trajectory being executed. Verify obstacle avoidance and (optionally) use a generated sample as a warm start for a collocation solver.

5 Part B — Simulation and Animation (30 points)

Choose **two of the four problems** and implement, simulate, and **animate** them. Any programming language is allowed (Python, MATLAB, Julia, C++, ...).

Difficulty-weighted scoring

- **Tier 1** — Problem 1 (iLQR/DDP) or Problem 2 (MPC): up to **13 pts** each.
- **Tier 2** — Problem 3 (RL) or Problem 4 (Generative): up to **15 pts** each.
- Total is capped at **30 pts**. Two Tier-2 selections can reach 30; two Tier-1 reach 26; a mix reaches 28. Choosing harder problems is rewarded.

Common deliverables (per simulated problem):

1. **Model:** the explicit discrete-time dynamics or ODE used for integration, *including* the system extension (actuator state / constraints / non-holonomic constraint / obstacles).
2. **Integrator:** name and justify it (RK4, `solve_ivp`, `ode45`, forward Euler, ...).
3. **Parameters and initial conditions** with justification.
4. **Animation (mandatory):** a rendered animation of the motion (cart sliding with swinging pole, the mobile inverted pendulum rolling/balancing, or the planar robot tracing a generated path). Acceptable forms: `.mp4`/`.gif`, Matplotlib `FuncAnimation`, MATLAB movie, or an interactive viewer. Include a still frame in the report and the path/link to the animation.
5. **At least one quantitative verification** appropriate to the method: iLQR cost-convergence and forward-rollout match (P1); constraint satisfaction and per-step solve time (P2); learning curve and baseline comparison (P3); obstacle-avoidance / multimodal coverage (P4).

6 Part C — Report Quality and Synthesis (10 points)

- Clear problem statements, complete derivations, and labeled figures.
- **Reproducibility:** provide code (repository link or appendix) with instructions to regenerate every plot and animation.
- **Cross-topic synthesis:** an explicit discussion connecting the four methods (e.g. iLQR as the inner solver of nonlinear MPC; RL learning value/policy where Riccati is unavailable; diffusion/flow models producing warm starts and multimodal candidates for the optimizers).
- Correct citations to the lecture notes and the listed references.

7 Project Presentation (30 points)

Prepare PowerPoint or PDF slides and **record a narrated video presentation that is strictly under 10 minutes**. Submit the slide file (`.pptx` or `.pdf`) and the recording as an `.mp4` inside the project `.zip`. The video should walk through all four problems at a high level and show the simulation animation(s) for your two selected problems.

Component	Points
Slides — clarity & structure (problem framing, method, results)	12
Technical depth & correctness (derivations/algorithms; honest limitations)	10
Animation / demo embedded in the video (at least one simulated system)	5
Recording quality & timing (clear narration, **under 10 min**)	3
Total	**30**

End of Final Project

Report 120 pts (Part A 80 + Part B 30 + Part C 10) + Presentation 30 pts = Total 150 points